

SIMATS ENGINEERING

ASSESSMENT TOOL 2 — ARTICLE REVIEW

Selected Article — Citation Details

Field	Details
Authors	Long Zhao, Sanghyuk Lee, Seon-Phil Jeong
Title	Decision Tree Application to Classification Problems with Boosting Algorithm
Journal	Electronics (MDPI), Volume 10, Issue 16, Article 1903
Year	2021 (Published: 8 August 2021)
DOI	https://doi.org/10.3390/electronics10161903
Domain / Application	Banking & Finance — Personal credit evaluation / term-deposit subscription prediction
ML Problem Addressed	Binary classification of bank customers (subscribe vs. not subscribe to a term deposit) using decision tree algorithms enhanced with a boosting technique

Task 1: Article Summary

(a) Citation, Domain and ML Problem

Full citation (APA style): Zhao, L., Lee, S., & Jeong, S.-P. (2021). Decision Tree Application to Classification Problems with Boosting Algorithm. *Electronics*, 10(16), 1903. <https://doi.org/10.3390/electronics10161903>

The article belongs to the banking and financial services domain, specifically personal credit evaluation. The specific machine learning problem addressed is a binary classification task: predicting whether a bank customer will subscribe to a term (fixed) deposit product, based on the customer's demographic, financial, and campaign-contact attributes. The authors approach this as a supervised inductive-learning problem solved using decision tree classifiers (C5.0 and CART) combined with a boosting ensemble technique.

(b) Key Objective and Research Gap (150–200 words)

The article's key objective is to design and evaluate decision-tree-based classifiers — C5.0 and CART — for predicting customer subscription to a bank term deposit, and to test whether adding a boosting algorithm and a cost matrix improves classification performance. The authors note that conventional C5.0 and CART trees often overfit or perform poorly on imbalanced datasets, where the number of "no" responses vastly outnumbers "yes" responses. This imbalance causes standard decision trees to become biased toward the majority class, producing high overall accuracy but poor sensitivity toward the minority (subscribing) class — the customers a bank actually wants to identify. The research gap the authors identify is the lack of a systematic comparison of how boosting and cost-sensitive learning (via a cost matrix) affect decision tree performance on real, imbalanced banking data,

evaluated using confusion-matrix metrics and 10-fold cross-validation rather than accuracy alone. They fill this gap by benchmarking five model variants (C5.0, C5.0+CM, C5.0+Boosting, CART, CART+Boosting) on the same dataset and analysing accuracy, precision, sensitivity, specificity, F1-score, Kappa, and fitting time together.

Task 2: Methodology Analysis

(a) ML Technique(s) and Dataset

The article uses two decision tree variants: (1) C5.0, which selects splitting attributes using information-gain ratio, and (2) CART (Classification and Regression Tree), which selects splits using the Gini coefficient. Both are combined with an AdaBoost-style boosting algorithm: multiple weak tree classifiers are trained sequentially, each round re-weighting misclassified samples, and the final strong classifier is a weighted vote of all weak learners ($H(x) = \text{sign}(\sum \alpha_t \cdot h_t(x))$). For C5.0, the authors additionally test a cost matrix (CM) that penalises misclassifying a true subscriber more heavily than the reverse error, to counteract class imbalance.

Dataset: The Bank Marketing dataset from a Portuguese bank, containing 45,211 customer records in total (40,690 used for model fitting/training and 4,521 held out as the test set). It has 17 variables — 7 continuous (age, balance, day, duration, campaign, pdays, previous) and 9 discrete (job, marital status, education, default, housing loan, personal loan, contact type, month, previous-campaign outcome) — plus the binary target variable y (subscribed to term deposit: yes/no). Before modelling, the authors checked Pearson correlation (all pairwise coefficients below 0.5) and Variance Inflation Factor (all VVIF < 2) to confirm no problematic multicollinearity, concluding that dimensionality reduction (e.g., PCA) was unnecessary.

(b) Mapping to CO4 Topics and Strength/Limitation

The technique maps directly onto two CO4 topics. First, Decision Trees: C5.0 and CART are classic decision-tree induction algorithms, and the paper demonstrates their practical construction (tree sizes of 414 and 74 nodes respectively) and evaluation on real data. Second, Inductive Learning: both algorithms learn general classification rules by generalising from 40,690 labelled training examples — the core inductive-learning paradigm from the syllabus. The boosting/ensemble mechanism can also be viewed as a Statistical Learning enhancement, since it iteratively reweights the training distribution to minimise empirical error.

Methodological strength: The study does not rely on accuracy alone. It uses a full confusion-matrix breakdown, multiple metrics (precision, sensitivity, specificity, F1, Kappa), and 10-fold cross-validation, giving a much more trustworthy picture of performance on an imbalanced dataset than a single accuracy figure would.

Methodological limitation: The dataset's severe class imbalance (far more "no" than "yes" customers) is never directly corrected through resampling (e.g., SMOTE, undersampling); the authors only compensate indirectly via a cost matrix on one model (C5.0). This leaves sensitivity for the minority "yes" class comparatively low (e.g., 0.4558 for plain C5.0) even though overall accuracy looks strong (~90%), which can be misleading if accuracy is read in isolation.

Task 3: Results & Critical Evaluation

(a) Key Results — Comparative Table

The authors benchmark five model variants on the held-out test set (Positive = 'yes', i.e., customer subscribes). Reported metrics from the article are summarised below:

Model	Accuracy	Precision	Sensitivity	Specificity	F1-Score	Fit Time
C5.0	90.56%	45.60%	63.24%	93.05%	53.05%	4 s
C5.0 + Cost Matrix	89.60%	70.32%	54.31%	95.91%	61.29%	4 s
C5.0 + Boosting	90.33%	45.18%	61.92%	92.99%	52.24%	8 s
CART	90.31%	37.24%	65.02%	92.13%	47.36%	10 s
CART + Boosting	90.95%	65.23%	48.58%	96.57%	55.69%	2 min

The average 10-fold cross-validation accuracy ranks the models as: CART + Boosting (90.7%) > C5.0 + Boosting (90.5%) > C5.0 (90.3%) > CART (90.2%) > C5.0 + Cost Matrix (90.1%) — all models performed above 90% average accuracy. CART + Boosting achieved the single highest test-set accuracy (90.95%) and precision improved most under boosting (CART's precision rose from 37.24% to 65.23%).

(b) Critical Evaluation of Results

The reported metrics are broadly realistic and internally consistent: the boosting variants require noticeably longer fitting time (CART + Boosting: ~2 minutes vs. 10 seconds for plain CART), which is an expected and credible trade-off rather than an inflated claim. However, a potential bias arises from the class-imbalanced evaluation protocol: because subscribers are a small minority, a model can reach ~90% accuracy while still misclassifying a large share of the customers who actually subscribe (sensitivity as low as 45–65% across models). This means accuracy alone overstates practical usefulness, and the paper's own Kappa values (0.48–0.55) confirm only moderate agreement beyond chance. Reported results also come from a single train/test split (40,690/4,521) supplemented by 10-fold cross-validation on the same dataset, so there is no external validation on a different bank or time period, which limits how far the results can be generalised.

Additional experiments that would strengthen the findings include: (i) applying class-balancing techniques such as SMOTE or class-weighted loss and comparing against the cost-matrix approach; (ii) testing on an independent, out-of-time dataset (e.g., a later campaign or a different bank) to check temporal/domain generalisation; (iii) reporting confidence intervals or statistical significance tests across the 10 cross-validation folds rather than only mean accuracy; and (iv) benchmarking against other ensemble methods (Random Forest, XGBoost, LightGBM) to contextualise how much of the improvement is attributable specifically to boosting versus ensembling in general.

(c) Real-World Applicability

This technique is directly deployable in retail-banking marketing analytics — for example, ranking customers by predicted likelihood of subscribing to a term deposit so call-centre agents can prioritise high-probability leads, improving campaign conversion rates and reducing wasted contact effort. Similar decision-tree-plus-boosting pipelines are also applicable to credit scoring, loan-default prediction, and churn prediction, since these all share the same tabular, mixed continuous/discrete feature structure.

Challenges in scaling to industry include: data availability and quality (real customer data requires strict privacy handling and may contain more noise/missing values than a curated benchmark dataset); compute and latency (the 2-minute fit time for CART + Boosting is trivial for offline batch scoring but could matter for real-time decisioning at scale); model drift (customer behaviour and macroeconomic conditions change, so the model needs periodic retraining/monitoring); and ethical/regulatory issues (using demographic attributes such as marital status or education in a credit-adjacent decision risks disparate impact and may require fairness auditing and regulatory compliance, e.g., under fair-lending laws).

Task 4: Reflection & Learning Outcome

(a) Three Key Insights Connected to CO4

- Insight 1 — Decision trees are interpretable but sensitive to split criteria: Comparing C5.0 (information-gain ratio) and CART (Gini coefficient) on the same data produced trees of very different sizes (414 vs. 74 nodes) and different attribute usage, directly illustrating the CO4 concept that the choice of splitting criterion changes both model complexity and generalisation, not just accuracy.
- Insight 2 — Ensemble/boosting is a statistical-learning enhancement, not a guaranteed improvement: Boosting improved CART noticeably (accuracy 90.31% → 90.95%, precision 37.24% → 65.23%) but barely changed C5.0's performance. This connects to the CO4 topic of statistical learning methods, showing that ensemble techniques interact with the base learner's existing bias/variance profile rather than universally boosting performance.
- Insight 3 — Accuracy alone is an inadequate evaluation criterion for imbalanced classification: All five models scored above 90% accuracy, yet sensitivity for the minority class ranged only from ~45% to ~65%. This reinforced the CO4 concept that a classifier's performance must be judged with confusion-matrix-derived metrics (precision, recall/sensitivity, F1, Kappa) appropriate to the class distribution and the real decision-making cost of each error type.

(b) Proposed Extension — Novel Experiment

Proposed enhancement: Apply a hybrid class-balancing + boosting pipeline — specifically, combine SMOTE (Synthetic Minority Over-sampling Technique) on the training set with a Gradient-Boosted Decision Tree (e.g., XGBoost or LightGBM) instead of the classical AdaBoost-style boosting used in the article, and evaluate on a newer, publicly available bank-marketing or micro-lending dataset (different institution/time period) to test generalisation.

Feasibility: This is highly feasible — SMOTE and gradient-boosting libraries (imbalanced-learn, XGBoost, LightGBM) are mature, open-source, and computationally efficient even on datasets of this size (~45,000 rows), so no specialised infrastructure is required.

Expected impact: Combining SMOTE with gradient boosting is expected to substantially raise sensitivity/recall for the minority "subscribe" class beyond the 45–65% range reported in the article, while keeping training time manageable (gradient-boosting implementations such as XGBoost are typically faster than the AdaBoost-style boosting used here). Validating on an independent dataset would also directly address the generalisation gap

identified in Task 3(b), giving a more industry-ready and CO4-aligned demonstration of how inductive learning, decision trees, and ensemble/statistical learning methods combine to solve a realistic classification and decision-making problem.
